

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/2024 Semester 1 MH1100 Calculus I

Tutorial 10, Week 11

Your tutor will aim to discuss: Problem 1, 2, 3, 4, 5, 6, 7, and 8

Problem 1 For function

$$f(x) = 2x^3 - 3x^2 - 12x$$

- (a) Find the intervals of increase or decrease;
- (b) Find the local maximum and minimum values;
- (c) Find the intervals of concavity and the inflection points.

[Solution:] Note that,

$$f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1)$$

- (a) so two critical points are $x = -1$ and $x = 2$.
when x is in $(-\infty, -1)$, $f'(x)$ is positive, so $f(x)$ increases in this region;
when x is in $(-1, 2)$, $f'(x)$ is negative, so $f(x)$ decreases in this region;
when x is in $(2, \infty)$, $f'(x)$ is positive, so $f(x)$ increases in this region.
- (b) We have

$$f''(x) = 12x - 6$$

- so $f''(-1) < 0$, $f(-1) = 7$ is local maximum value,
and $f''(2) > 0$, $f(2) = -20$ is local minimum value.
- (c) From the above results, we have
 $f''(x) < 0$ when x is in $(-\infty, 0.5)$, function is concave downward in this region.
 $f''(x) > 0$ when x is in $(0.5, \infty)$, function is concave upward in this region.
Point $(0.5, -6.5)$ is the inflection point.

Problem 2 Prove that $\tan x > x$ for $0 < x < \pi/2$.

[Solution:] Let

$$f(x) = \tan x - x$$

we have $f(0) = 0$, and when $0 < x < \pi/2$

$$f'(x) = \sec^2 x - 1 > 0$$

therefore $f(x)$ increases in the region $0 < x < \pi/2$, so $f(x) > f(0)$, and we have $\tan x > x$.

Problem 3 Compute the limit

(a)

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x)$$

(b)

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x}$$

1

2

(c)

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 7}{5x^2 + x - 3}$$

[Solution:] (a) We can have,

$$\begin{aligned} \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) &= \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) \cdot \frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{(x^2 + 1) - x^2}{\sqrt{x^2 + 1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x} \\ &= 0 \end{aligned}$$

(b) We let $t = 1/x$

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \lim_{t \rightarrow 0^+} \sin t = 0$$

(c) We have

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2x^2 - 7}{5x^2 + x - 3} &= \lim_{x \rightarrow \infty} \frac{2 - 7/x^2}{5 + 1/x - 3/x^2} \\ &= \frac{\lim_{x \rightarrow \infty} 2 - \lim_{x \rightarrow \infty} 7/x^2}{\lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} 1/x - \lim_{x \rightarrow \infty} 3/x^2} \\ &= \frac{2}{5} \end{aligned}$$

Problem 4 Calculate

$$\lim_{x \rightarrow \infty} \sqrt{\frac{12x^3 - 5x + 2}{1 + 4x^2 + 3x^3}}$$

[Solution:] We have

$$\begin{aligned} \lim_{x \rightarrow \infty} \sqrt{\frac{12x^3 - 5x + 2}{1 + 4x^2 + 3x^3}} &= \lim_{x \rightarrow \infty} \sqrt{\frac{12 - 5/x^2 + 2/x^3}{1/x^3 + 4/x + 3}} \\ &= \sqrt{\frac{\lim_{x \rightarrow \infty} 12 - \lim_{x \rightarrow \infty} 5/x^2 + \lim_{x \rightarrow \infty} 2/x^3}{\lim_{x \rightarrow \infty} 1/x^3 + \lim_{x \rightarrow \infty} 4/x + \lim_{x \rightarrow \infty} 3}} \\ &= \sqrt{\frac{12}{3}} \\ &= 2 \end{aligned}$$

Problem 5 Find the horizontal asymptotes of the curve

$$f(x) = \sqrt{3x^2 + 8x + 6} - \sqrt{3x^2 + 3x + 1}$$

[Solution:] Notice that

$$\begin{aligned}
& \lim_{x \rightarrow \infty} \sqrt{3x^2 + 8x + 6} - \sqrt{3x^2 + 3x + 1} \\
&= \lim_{x \rightarrow \infty} \frac{(3x^2 + 8x + 6) - (3x^2 + 3x + 1)}{\sqrt{3x^2 + 8x + 6} + \sqrt{3x^2 + 3x + 1}} \\
&= \lim_{x \rightarrow \infty} \frac{5x + 5}{\sqrt{3x^2 + 8x + 6} + \sqrt{3x^2 + 3x + 1}} \\
&= \lim_{x \rightarrow \infty} \frac{5 + 5/x}{\sqrt{3 + 8/x + 6/x^2} + \sqrt{3 + 3/x + 1/x^2}} \\
&= \frac{5}{2\sqrt{3}}
\end{aligned}$$

Further we have

$$\begin{aligned}
& \lim_{x \rightarrow -\infty} \sqrt{3x^2 + 8x + 6} - \sqrt{3x^2 + 3x + 1} \\
&= \lim_{x \rightarrow -\infty} \frac{(3x^2 + 8x + 6) - (3x^2 + 3x + 1)}{\sqrt{3x^2 + 8x + 6} + \sqrt{3x^2 + 3x + 1}} \\
&= \lim_{x \rightarrow -\infty} \frac{5x + 5}{\sqrt{3x^2 + 8x + 6} + \sqrt{3x^2 + 3x + 1}} \\
&= \lim_{x \rightarrow -\infty} - \frac{5 + 5/x}{\sqrt{3 + 8/x + 6/x^2} + \sqrt{3 + 3/x + 1/x^2}} \\
&= -\frac{5}{2\sqrt{3}}
\end{aligned}$$

Therefore $y = \frac{5}{2\sqrt{3}}$ and $y = -\frac{5}{2\sqrt{3}}$ are two horizontal asymptotes.

Problem 6 Find the limit or show it does not exist

(a)

$$\lim_{x \rightarrow \infty} x \sin(1/x)$$

(b)

$$\lim_{x \rightarrow \infty} \sqrt{x} \sin(1/x)$$

(c)

$$\lim_{x \rightarrow \infty} x^5 - x^4$$

[Solution:]

(a) We let $t = 1/x$

$$\lim_{x \rightarrow \infty} x \sin(1/x) = \lim_{t \rightarrow 0^+} \frac{\sin t}{t} = 1$$

(b) We let $t = 1/x$

$$\lim_{x \rightarrow \infty} \sqrt{x} \sin(1/x) = \lim_{t \rightarrow 0^+} \frac{\sin t}{t^{1/2}} = \lim_{t \rightarrow 0^+} \frac{\sin t}{t} t^{1/2} = 0$$

(c) We have

$$\lim_{x \rightarrow \infty} x^5 - x^4 = \lim_{x \rightarrow \infty} x^4(x - 1) = \infty$$

Problem 7 Let $P(x)$ and $Q(x)$ be polynomials. Find

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)}$$

- (a) if the degree of $P(x)$ is larger than $Q(x)$;
 (b) if the degree of $Q(x)$ is larger than $P(x)$.

[Solution:]

Divide the numerator and denominator by the highest power of x in $Q(x)$.

- (a) When $\deg P(x) > \deg Q(x)$, then the numerator goes to $\pm\infty$, but the denominator doesn't. So

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)} = \pm\infty$$

- (b) When $\deg P(x) < \deg Q(x)$, then the numerator goes to zero, but the denominator doesn't. So

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)} = 0$$

Problem 8 A tank contains 5000L pure water. We pump the salt water, which contains 30g/L concentration of salt, into the tank at a rate of 25L/min.

- (a) Show that the concentration of salt (g/L) after t minutes is

$$C(t) = \frac{30t}{200 + t}$$

- (b) What happens to the concentration as $t \rightarrow \infty$?

[Solution:]

- (a) After t min, the concentration will be

$$C(t) = \frac{30 \cdot 25 \cdot t}{5000 + 25 \cdot t} = \frac{30t}{200 + t} \text{ (g/L)}$$

- (b) We take the limit

$$\lim_{t \rightarrow \infty} C(t) = \lim_{t \rightarrow \infty} \frac{30t}{200 + t} = \lim_{t \rightarrow \infty} \frac{30}{200/t + 1} = 30 \text{ (g/L)}$$

The concentration will approach to 30g/L when $t \rightarrow \infty$.

Problem 9 Show that the equation

$$2x - 1 - \sin x = 0$$

has exactly one real root.

[Solution:]

Let $f(x) = 2x - 1 - \sin x$. Then $f(0) = -1 < 0$ and $f(\frac{\pi}{2}) = \pi - 2 > 0$. Since $f(x)$ is continuous, from the Intermediate Value Theorem, there is a number $c \in (0, \pi/2)$, such that $f(c) = 0$. Thus the equation has at least one real root.

If the equation has two distinct real roots, say a and b and $a < b$, then $f(a) = f(b) = 0$. Since $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , from Rolle's Theorem, there

exists $r \in (a, b)$, such that $f'(r) = 0$. However $f'(x) = 2 - \cos x > 0$, this contradiction shows the equation can not have two distinct real roots.

Problem 10 If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ possibly be?

[Solution:]

From the Mean Value Theorem, there is a $c \in (1, 4)$, such that $f(4) - f(1) = f'(c)(4 - 1)$. Since we have $f(1) = 10$ and $f'(c) \geq 2$ ($c \in (1, 4)$), we have

$$f(4) = f(1) + f'(c)(4 - 1) = 10 + 3f'(c) \geq 10 + 3 \cdot 2 = 16$$

So the smallest possible value of $f(4)$ is 16.